A MICRO PROJECT REPORT

ON

"A CAREER RECOMMENDATION SYSTEM"

Subject: Foundation of Artificial Intelligence

(Course Code: 2ADPC204)

Submitted in partial fulfilment of the requirement for

The award of the Degree in SY (AI & DS).

SUBMITTED BY

1. Name of Student :- **Avanti Ganesh Dhumale.** Roll No.:- **2034**

UNDER THE GUIDANCE OF

Prof. Ms.Priyanka Suryawanshi.

Annasaheb Dange College of Engineering and Technology,Ashta

Department of AI & DS.

Academic Year: 2026 – 2027.

CERTIFICATE

This is to certify that Ms. **Avanti Dhumale.** of III Semester of Degree in SY B. Tech of Institute ADCET has successfully completed the Micro Project work entitled " **AICAREER RECOMMENDATION System** " in the subject **" Foundation Of AI& DS**"(Course Code: 2ADPC204) for the academic year 2026 to 2027 as prescribed in the curriculum.

Place: Ashta

Date: _____

Subject Teacher
(Signature)

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my guide Prof. **Ms.Priyanka Suryawanshi** for the valuable guidance and constant support throughout this micro project on **"AI Career Recommendation System**" I am also thankful to **Mrs. Asma Shaikh,** Head of the Department, and to our Director **Dr. L. Y. Waghmode**. for providing the computer laboratory and software facilities required to complete this work. Finally, I thank all the teaching and non-teaching staff, my family and my friends who helped me directly or indirectly in completing this project.

Avanti Dhumale.
Roll No:- 2034

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MICRO PROJECT REPORT

AI CAREER RECOMMENDATION SYSTEM

Aims / Benefits of the Micro Project

Aim:

To develop an AI Career Recommendation Website that helps students and beginners explore suitable career options based on their interests, skills, and career goals.

Benefits:

1. It helps students understand different career opportunities.
2. It provides guidance for selecting a suitable career path.
3. It helps users identify the skills required for different careers.
4. It reduces confusion while choosing a career.
5. It creates awareness about opportunities in the field of Artificial Intelligence and other technical domains.
6. It provides career-related information through a simple and user-friendly website.
7. It encourages students to plan their learning and professional development.

Course Outcomes Achieved

1.Understood the basic concepts of Artificial Intelligence.

2. Learned how technology can be used to solve real-world problems.

3. Understood the importance of career recommendation systems.

4. Learned the basic process of designing and developing a website.

5. Improved logical thinking and problem-solving skills.

6. Understood how user requirements can be used to provide relevant career guidance.

7. Gained practical knowledge of web-based applications.

Literature Review

1.Career guidance is important for students because it helps them understand their interests, strengths, and future opportunities. Traditional career guidance is usually provided by teachers, counsellors, websites, and educational institutions.

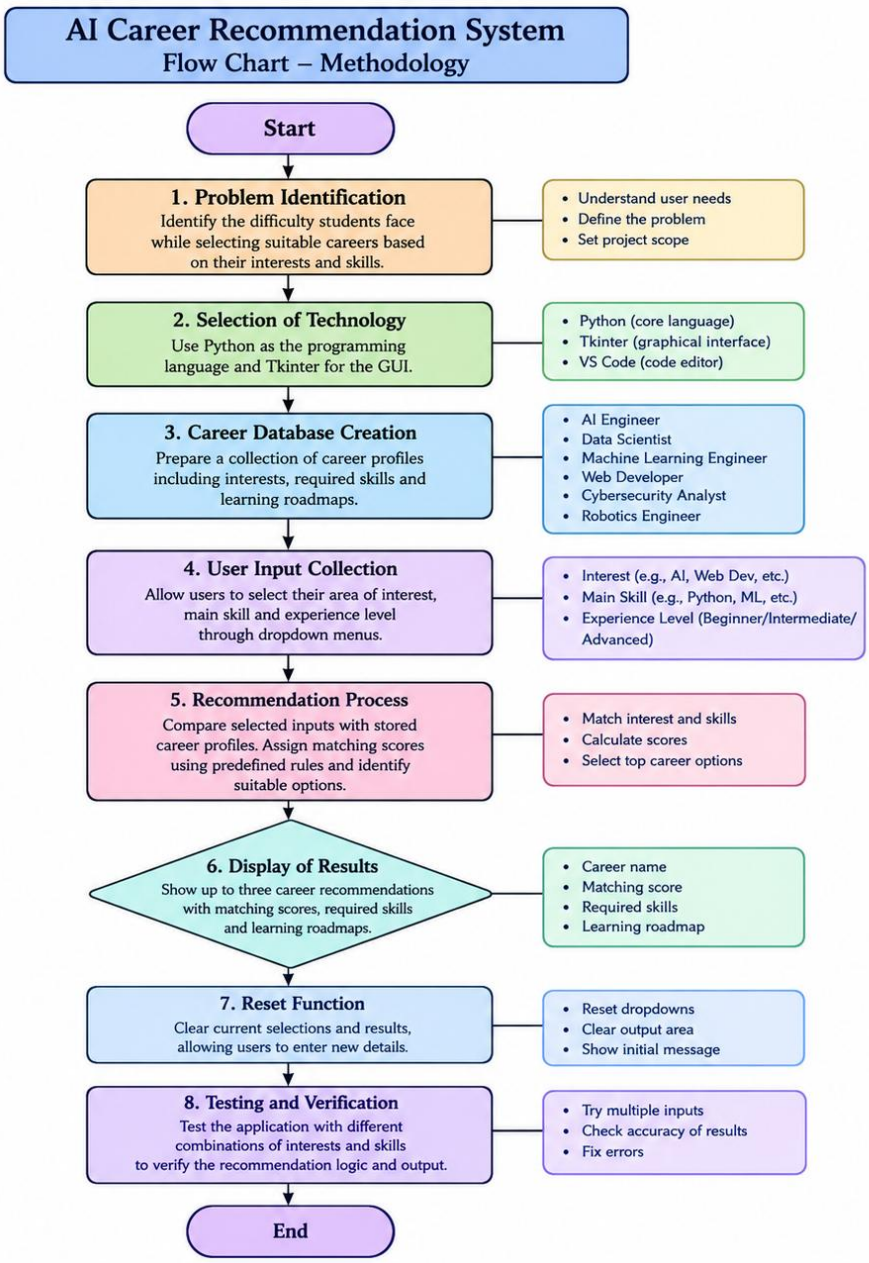
2.With the development of Artificial Intelligence, computer-based recommendation systems can help users explore suitable options based on their preferences. Recommendation systems are widely used in education, online shopping, entertainment, and other fields.

3.Python is a popular programming language because of its simple syntax and wide range of applications. Tkinter is a Python library used to create graphical user interfaces for desktop applications.

4.In this project, Python is used to store career information and implement the recommendation logic. Tkinter provides the graphical interface through which users select their interests and skills.

5.The system compares the selected inputs with predefined career profiles and displays the most relevant matches. This project uses rule-based recommendation logic rather than a trained machine learning model. It is designed as a basic prototype to demonstrate how a recommendation system can work.

Actual Methodology Followed



Actual Resources Used

Hardware:

- Computer or laptop
- Keyboard and monitor

Software:

- Operating System: Windows
- Programming Language: Python
- Code Editor: Visual Studio Code (VS Code)

Other Resources:

- Python and Tkinter documentation
- Course notes on Artificial Intelligence
- Career-related information
- Internet resources for reference and learning

Outputs of the Micro Project

Welcome | ai_career_recommendation.py

ai_career_recommendation.py > ...

```python

import tkinter as tk
from tkinter import ttk, messagebox

# Career information and required skills
CAREERS = {
    "AI Engineer": {
        "interests": ["Artificial Intelligence", "Programming", "Mathematics"],
        "skills": ["Python", "Machine Learning", "Mathematics"],
        "roadmap": "Learn Python, Mathematics, Machine Learning and AI projects."
    },
    "Data Scientist": {
        "interests": ["Data Analysis", "Mathematics", "Problem Solving"],
        "skills": ["Python", "Statistics", "Data Analysis"],
        "roadmap": "Learn Python, Statistics, Pandas, Data Visualization and ML."
    },
    "Machine Learning Engineer": {
        "interests": ["Artificial Intelligence", "Programming", "Mathematics"],
        "skills": ["Python", "Machine Learning", "Mathematics"],
        "roadmap": "Learn Python, ML algorithms, model evaluation and deployment."
    },
    "Web Developer": {
        "interests": ["Web Development", "Programming", "Design"],
        "skills": ["HTML/CSS", "JavaScript", "Python"],
        "roadmap": "Learn HTML, CSS, JavaScript and build practical websites."
    },
    "Cybersecurity Analyst": {
        "interests": ["Cybersecurity", "Problem Solving", "Programming"],
        "skills": ["Networking", "Python", "Problem Solving"],
        "roadmap": "Learn networking, Linux, security fundamentals and ethical testing."
    },
    "Robotics Engineer": {
        "interests": ["Robotics", "Mathematics", "Programming"],
        "skills": ["Python", "Mathematics", "Problem Solving"],
        "roadmap": "Learn Python, electronics, sensors, control systems and robotics."
    }
}


class CareerRecommendationApp:
    def __init__(self, root):
        self.root = root
        self.root.title("AI Career Recommendation System")
        self.root.geometry("760x700")
        self.root.configure(bg="#eef3f9")
        self.root.minsize(650, 620)

        title = tk.Label(
            root,
            text="AI CAREER COMPASS",
            font=("Arial", 22, "bold"),
            bg="#183153",
            fg="white",
            pady=16
        )
        title.pack(fill="x")

        subtitle = tk.Label(
            root,
            text="Discover a career that matches your interests and skills",
            font=("Arial", 11),
            bg="#eef3f9",
            fg="#183153",
            pady=10
        )
        subtitle.pack()
```

Welcome ai_career_recommendation.py

ai_career_recommendation.py > ...

```python
class CareerRecommendationApp:
    def __init__(self, root):
        )
        subtitle.pack()

        form = tk.Frame(root, bg="white", padx=20, pady=18)
        form.pack(fill="x", padx=25, pady=8)

        tk.Label(
            form, text="Select Your Interest:",
            font=("Arial", 11, "bold"), bg="white"
        ).grid(row=0, column=0, sticky="w", pady=10)

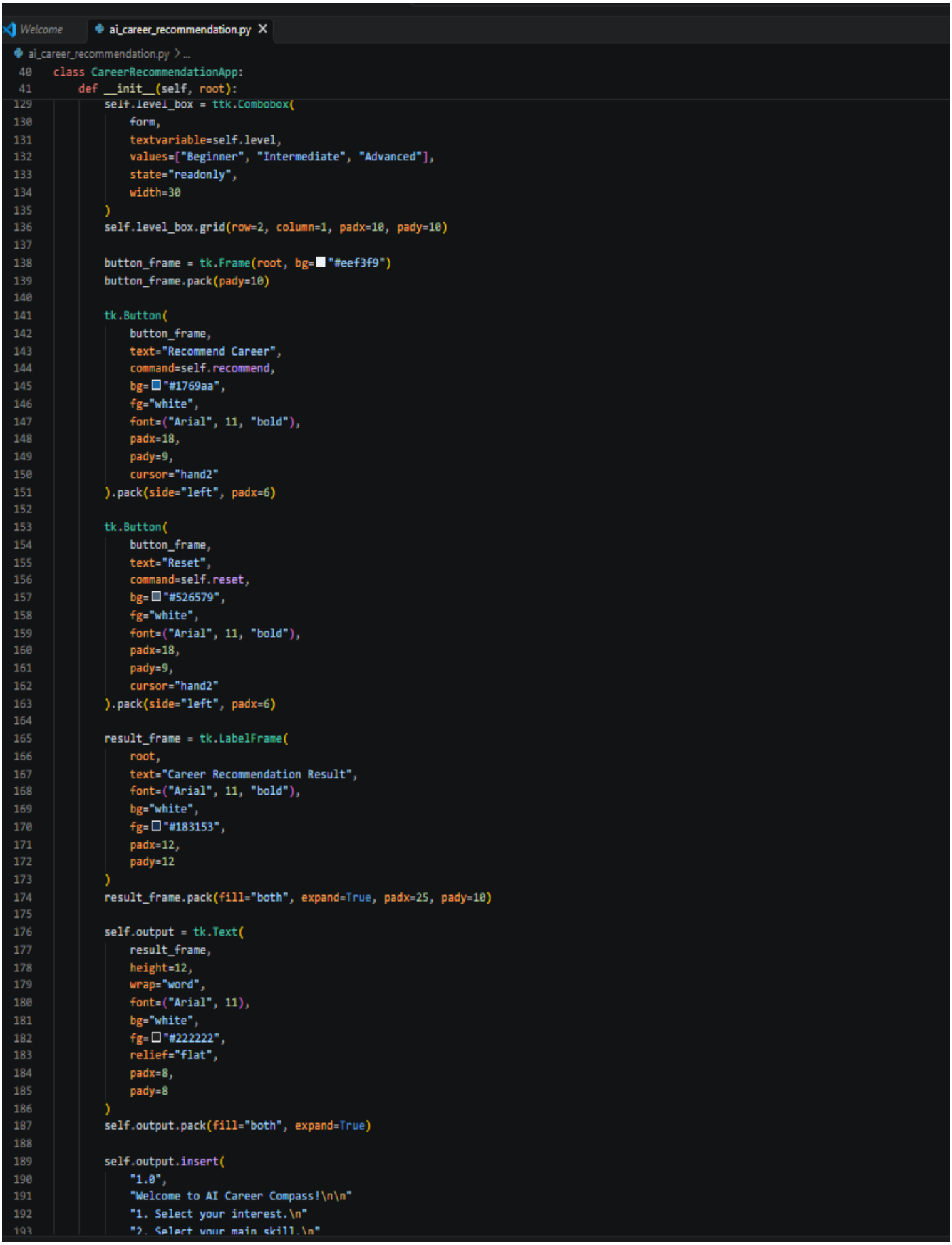
        self.interest = tk.StringVar()
        self.interest_box = ttk.Combobox(
            form,
            textvariable=self.interest,
            values=[
                "Artificial Intelligence",
                "Data Analysis",
                "Web Development",
                "Cybersecurity",
                "Robotics",
                "Programming",
                "Mathematics",
                "Problem Solving",
                "Design"
            ],
            state="readonly",
            width=30
        )
        self.interest_box.grid(row=0, column=1, padx=10, pady=10)
        self.interest_box.current(0)

        tk.Label(
            form, text="Select Your Main Skill:",
            font=("Arial", 11, "bold"), bg="white"
        ).grid(row=1, column=0, sticky="w", pady=10)

        self.skill = tk.StringVar()
        self.skill_box = ttk.Combobox(
            form,
            textvariable=self.skill,
            values=[
                "Python",
                "Machine Learning",
                "Mathematics",
                "Statistics",
                "Data Analysis",
                "HTML/CSS",
                "JavaScript",
                "Networking",
                "Problem Solving"
            ],
            state="readonly",
            width=30
        )
        self.skill_box.grid(row=1, column=1, padx=10, pady=10)
        self.skill_box.current(0)

        tk.Label(
            form, text="Experience Level:",
            font=("Arial", 11, "bold"), bg="white"
        ).grid(row=2, column=0, sticky="w", pady=10)

        self.level = tk.StringVar(value="Beginner")
        self.level_box = ttk.Combobox(
```


Welcome
ai_career_recommendation.py
ai_career_recommendation.py > ...
40 class CareerRecommendationApp:
41 def __init__(self, root):
129 self.level_box = ttk.Combobox(
130 form,
131 textvariable=self.level,
132 values=["Beginner", "Intermediate", "Advanced"],
133 state="readonly",
134 width=30
135)
136 self.level_box.grid(row=2, column=1, padx=10, pady=10)
137
138 button_frame = tk.Frame(root, bg="#eef3f9")
139 button_frame.pack(pady=10)
140
141 tk.Button(
142 button_frame,
143 text="Recommend Career",
144 command=self.recommend,
145 bg="#1769aa",
146 fg="white",
147 font=("Arial", 11, "bold"),
148 padx=18,
149 pady=9,
150 cursor="hand2"
151).pack(side="left", padx=6)
152
153 tk.Button(
154 button_frame,
155 text="Reset",
156 command=self.reset,
157 bg="#526579",
158 fg="white",
159 font=("Arial", 11, "bold"),
160 padx=18,
161 pady=9,
162 cursor="hand2"
163).pack(side="left", padx=6)
164
165 result_frame = tk.LabelFrame(
166 root,
167 text="Career Recommendation Result",
168 font=("Arial", 11, "bold"),
169 bg="white",
170 fg="#183153",
171 padx=12,
172 pady=12
173)
174 result_frame.pack(fill="both", expand=True, padx=25, pady=10)
175
176 self.output = tk.Text(
177 result_frame,
178 height=12,
179 wrap="word",
180 font=("Arial", 11),
181 bg="white",
182 fg="#222222",
183 relief="flat",
184 padx=8,
185 pady=8
186)
187 self.output.pack(fill="both", expand=True)
188
189 self.output.insert(
190 "1.0",
191 "Welcome to AI Career Compass!\n\n"
192 "1. Select your interest.\n"
193 "2. Select your main skill.\n"

Welcome | ai_career_recommendation.py

ai_career_recommendation.py > ...

```
class CareerRecommendationApp:
    def __init__(self, root):
            "2. Select your main skill.\n"
            "3. Choose your experience level.\n"
            "4. Click 'Recommend Career' to view suggestions.\n"
        )
        self.output.config(state="disabled")

    def recommend(self):
        interest = self.interest.get()
        skill = self.skill.get()
        level = self.level.get()

        if not interest or not skill or not level:
            messagebox.showwarning(
                "Missing Information",
                "Please select all the fields."
            )
            return

        results = []

        for career, details in CAREERS.items():
            score = 0

            if interest in details["interests"]:
                score += 2

            if skill in details["skills"]:
                score += 2

            # Give a small bonus for problem-solving interests
            if interest == "Problem Solving":
                if "Problem Solving" in details["skills"]:
                    score += 1

            if score > 0:
                results.append((career, score, details))

        results.sort(key=lambda item: item[1], reverse=True)

        self.output.config(state="normal")
        self.output.delete("1.0", tk.END)

        self.output.insert(
            tk.END,
            "YOUR CAREER RECOMMENDATION\n"
            "==================================\n\n"
        )
        self.output.insert(tk.END, f"Interest: {interest}\n")
        self.output.insert(tk.END, f"Main Skill: {skill}\n")
        self.output.insert(tk.END, f"Experience: {level}\n\n")

        if not results:
            self.output.insert(
                tk.END,
                "No strong match found.\n"
                "Try another interest or skill combination.\n"
            )
        else:
            self.output.insert(
                tk.END,
                "TOP MATCHING CAREER OPTIONS\n\n"
            )
```

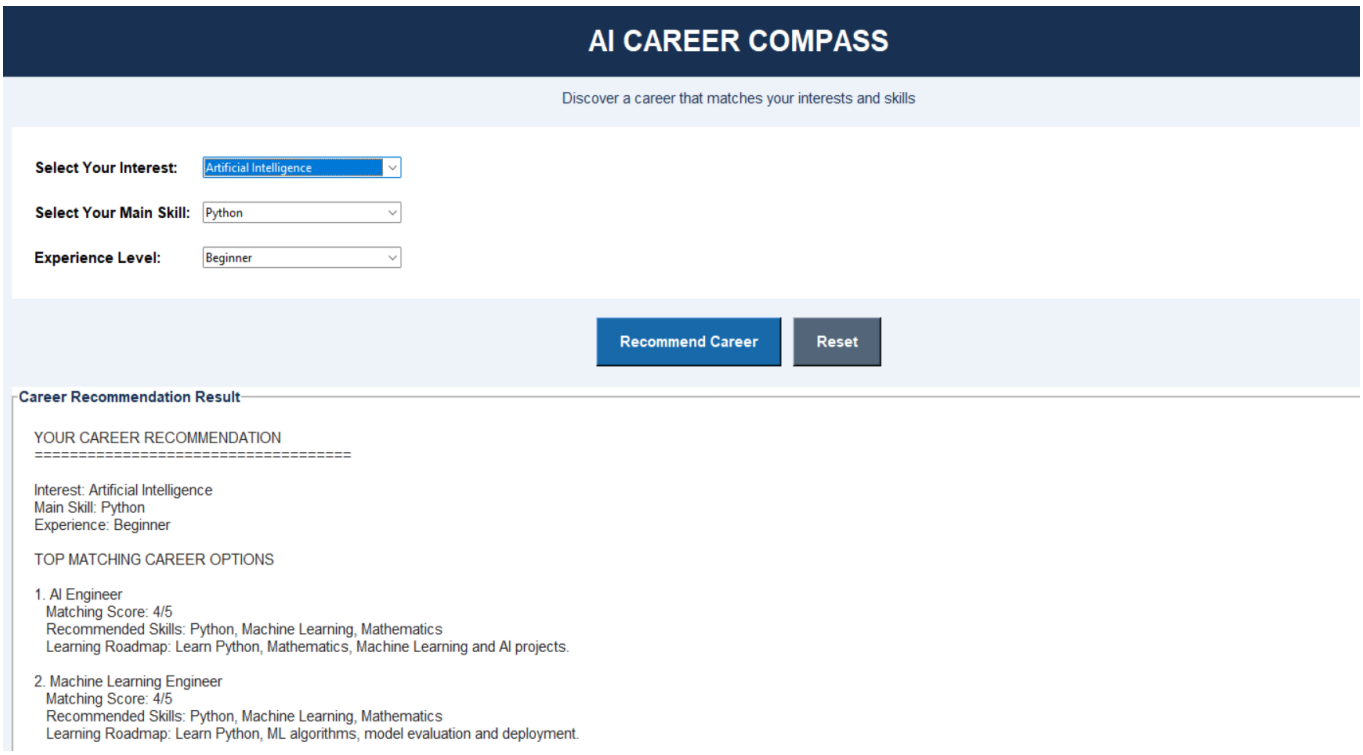
```
                results[:3], start=1
            ):
                self.output.insert(
                    tk.END,
                    f"{index}. {career}\n"
                    f"   Matching Score: {score}/5\n"
                    f"   Recommended Skills: "
                    f"{', '.join(details['skills'])}\n"
                    f"   Learning Roadmap: {details['roadmap']}\n\n"
                )

        self.output.insert(
            tk.END,
            "Note: These are basic rule-based suggestions, "
            "not a guaranteed career prediction."
        )
        self.output.config(state="disabled")

    def reset(self):
        self.interest_box.current(0)
        self.skill_box.current(0)
        self.level_box.set("Beginner")

        self.output.config(state="normal")
        self.output.delete("1.0", tk.END)
        self.output.insert(
            "1.0",
            "Selections reset successfully.\n"
            "Choose your interest and skills to begin again."
        )
        self.output.config(state="disabled")


if __name__ == "__main__":
    root = tk.Tk()
    app = CareerRecommendationApp(root)
    root.mainloop()
```



AI CAREER COMPASS

Discover a career that matches your interests and skills

Select Your Interest: Data Analysis

Select Your Main Skill: Python

Experience Level: Beginner

Recommend Career | Reset

Career Recommendation Result

YOUR CAREER RECOMMENDATION
==========

Interest: Data Analysis
Main Skill: Python
Experience: Beginner

TOP MATCHING CAREER OPTIONS

1. Data Scientist
   Matching Score: 4/5
   Recommended Skills: Python, Statistics, Data Analysis
   Learning Roadmap: Learn Python, Statistics, Pandas, Data Visualization and ML.

2. AI Engineer
   Matching Score: 2/5
   Recommended Skills: Python, Machine Learning, Mathematics
   Learning Roadmap: Learn Python, Mathematics, Machine Learning and AI projects.

AI CAREER COMPASS

Discover a career that matches your interests and skills

Select Your Interest: Design

Select Your Main Skill: HTML/CSS

Experience Level: Advanced

Recommend Career | Reset

Career Recommendation Result

YOUR CAREER RECOMMENDATION
==========

Interest: Design
Main Skill: HTML/CSS
Experience: Advanced

TOP MATCHING CAREER OPTIONS

1. Web Developer
   Matching Score: 4/5
   Recommended Skills: HTML/CSS, JavaScript, Python
   Learning Roadmap: Learn HTML, CSS, JavaScript and build practical websites.

Note: These are basic rule-based suggestions, not a guaranteed career prediction.

Skills Developed

- Python programming.
- Logical thinking.
- Problem-solving.
- Error detection and debugging.
- Software development.
- Basic understanding of AI.

Applications of the Micro Project

AI Career Recommendation System can be used in the following areas:

1. **Student Career Exploration:** Helps students explore career options in technology and Artificial Intelligence.
2. **Educational Institutions:** Can be used as a basic career-awareness tool in colleges.
3. **Learning Path Planning:** Provides information about the skills required for different careers.
4. **Skill Development:** Helps users identify technical areas they may want to learn.
5. **Career Guidance Demonstration:** Demonstrates the basic working of a recommendation system.
6. **Future AI Applications:** Can be expanded into a more advanced system with a larger career database and machine learning techniques

Conclusion :-

- The AI Career Recommendation System was successfully designed as a basic desktop application using Python and Tkinter. It provides a user-friendly interface for selecting interests and technical skills and displays matching career options with required skills and learning roadmaps.
- The project demonstrates the practical application of Python programming, graphical user interface development, data organization, and rule-based recommendation logic. It also helped improve logical thinking, problem-solving, testing, and debugging skills.
- The current system provides basic career suggestions and does not replace professional career counselling. In the future, it can be improved by adding more career options, detailed learning resources, user profiles, and a trained machine learning model for more personalized recommendations.

Reference :-

- Python Official Documentation — https://docs.python.org/3/
- Python Tkinter Documentation — https://docs.python.org/3/library/tkinter.html
- Foundation of Artificial Intelligence course notes.
- Reference materials related to Python programming and career guidance.